

DATE:

EXPERIMENT NO:

AIM: Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

SCRIPT:

```
\documentclass{article} % Use the standard article class
\usepackage[a4paper, margin=1in]{geometry} % Set page size and margins
\usepackage{multicol} % Allows multiple columns in the document
\usepackage{fancyhdr} % Allows headers and footers
\begin{document} % Begin document content
\pagestyle{fancy} % For page headers/footers
\fancyhff % Clear default header and footer fields
\fancyhead[L]{2020 IEEE International Conference on Artificial Intelligence and
Computer Applications (ICAICA)}
\begin{center} % Center the title
\textbf{\LARGE Research on Expert System in Power Network Operation Ticket}
\end{center}
\setlength{\columnsep}{8pt} % Set spacing between columns
\begin{multicols}{3} % Set 3-column layout
\begin{center} % First author block centered
\textbf{Yuhang Li}||
Nari Technology (State Grid Electric Power Research Institute) Co., Ltd.|| Nanjing,
China|| Nari-Tech Power Distribution \& Rural Electrification Branch Company||
Nanjing, China
\end{center}
\columnbreak
\begin{center} % Second author
\textbf{Chenxi Huang}||
Nari Technology (State Grid Electric Power Research Institute) Co., Ltd.||
Nanjing, China||
Nari-Tech Power Distribution \& Rural Electrification Branch Company||
Nanjing, China
\end{center}
\columnbreak
\begin{center} % Third author
\textbf{Chao Lu}|| Nari Technology (State Grid Electric Power Research Institute)
Co., Ltd.||
Nanjing, China|| Nari-Tech Power Distribution \& Rural Electrification Branch
Company|| Nanjing, China
\end{center}
\end{multicols}
\begin{abstract}
\textbf{\textit{Abstract-}} This paper realizes the automatic generation system of
reversing operation ticket, aiming at the problem of insufficient flexibility and
intelligence by expert system technology. By analyzing the characteristics of the
power system, the knowledge base and the inference engine which constitute the
```

operation ticket expert system are studied separately to design a knowledge base with a more concise knowledge representation. The Rete algorithm was used in the study of reasoning mechanism, and the design of the operation ticket expert system was completed finally.

`\textit{Keywords-}` artificial intelligence, operation ticket, expert system, rete algorithm

`\end{abstract}`

`\end{document}`

AIM: Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry].

SCRIPT:

```
\documentclass[12pt,a4paper]{article}
\usepackage[a4paper,margin=1in]{geometry} \usepackage{setspace}
\usepackage{fancyhdr} \usepackage{ragged2e} \usepackage{times}
\setstretch{1.25} \pagestyle{fancy}
\fancyhf{}
\fancyfoot[C]{\thepage}
\fancyhead[L]{\textbf{Guru Nanak Dev Engineering College, Ludhiana}}
\fancyhead[R]{\textbf{Department of CSE}}
\begin{document}
\begin{titlepage}
\begin{center}
{\Large \textbf{Guru Nanak Dev Engineering College, Ludhiana}}\\[0.2cm]
{\large (An Autonomous College under UGC Act)}\\[0.3cm]
\hrule{\textwidth}{0.6pt}\\[1cm]
{\LARGE \textbf{CERTIFICATE}}\\[1.2cm]
\noindent
\begin{minipage}{0.9\textwidth}
\justifying
This is to certify that the project/report entitled \textbf{"\underline{\hspace{6cm}}"},
\textbf{\underline{\hspace{5.5cm}}}, submitted by \textbf{Branch: Computer Science
and Engineering}, bearing University Roll No. \textbf{\underline{\hspace{4cm}}}, in
partial fulfillment of the requirements for the award of the degree of
\textbf{\underline{\hspace{6cm}}} to \textbf{Guru Nanak Dev Engineering College
(Autonomous), Ludhiana}, is a bonafide record of the work carried out by him/her
under my supervision and guidance during the academic year
\textbf{\underline{\hspace{4cm}}}.
\end{minipage}
\vspace{2cm}
\begin{minipage}{0.47\textwidth}
\centering
\textbf{Signature of the Guide}\\[1cm]
\begin{tabular}{@{}l@{}}
Name: \underline{\hspace{6cm}} \\[0.5cm]
Designation: \underline{\hspace{5cm}} \\[0.5cm]
Department: \underline{\hspace{5cm}} \\
\end{tabular}
\end{minipage}
\hfill
\begin{minipage}{0.47\textwidth}
\centering
\textbf{Signature of the Head of Department}\\[1cm]
\begin{tabular}{@{}l@{}}
Name: \underline{\hspace{6cm}} \\[0.5cm]
Designation: \underline{\hspace{5cm}} \\[0.5cm]
\end{tabular}
\end{minipage}
\end{center}
\end{titlepage}
\end{document}
```

```

Department: \underline{\hspace{5cm}} \\
\end{tabular}
\end{minipage}
\vfill
\noindent
\begin{minipage}{0.45\textwidth}
\textbf{Date:} \underline{\hspace{5cm}}
\end{minipage}
\hfill
\begin{minipage}{0.45\textwidth}
\raggedleft \textbf{Place:} \underline{\hspace{5cm}}
\end{minipage}
\end{center}
\end{titlepage}
\end{document}

```

DATE: 08.09.2025

EXPERIMENT NO: 7

AIM: Develop a LaTeX script to create a document that contains the following table with proper labels:

S No.	URN	Student Name	Marks		
			Subject 1	Subject 2	Subject 3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

SCRIPT:

```
\documentclass[a4paper,12pt]{article}
\usepackage{geometry} \usepackage{fancyhdr}
\usepackage{caption} \usepackage{multirow}
\usepackage{array} \usepackage{booktabs}
\geometry{margin=1in}
\pagestyle{fancy}
\fancyhf{} % Clear default headers/footers
\fancyhead[L]{Experiment 07}
\fancyfoot[l]{GNDEC}
\fancyfoot[R]{URN: 2435157}
\renewcommand{\headrulewidth}{1pt}
\renewcommand{\footrulewidth}{1pt}
\title{\textbf{Student Record}}
\author{SAKSHAM}
\date{}
\begin{document}
\maketitle
\thispagestyle{fancy}
\noindent \large
\vspace{1em}
\begin{table}[h!]
\centering
\caption{Marks Obtained by Students in Three Subjects}
\renewcommand{\arraystretch}{1.3}
\begin{tabular}{|c|c|c|c|c|}
\hline
\multirow{2}{*}{\textbf{S.No}} &
\multirow{2}{*}{\textbf{URN}} &
\multirow{2}{*}{\textbf{Student Name}} &
\multicolumn{2}{\textbf{Marks}} \\
\hline
& & & \textbf{Data Structure} & \textbf{OOPs} & \textbf{Maths III} \\
\hline
1 & 243518501 & Simran & 89 & 60 & 90 \\
\hline
2 & 243518502 & Aman & 78 & 45 & 98 \\
\hline
\end{tabular}
\end{table}
```

```
3 & 243518503 & Divya & 67 & 55 & 59 \\  
\hline  
\end{tabular}  
\end{table}  
\end{document}
```

AIM: Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

SCRIPT:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc} \usepackage{graphicx} \usepackage{subcaption}
\usepackage{fancyhdr}
\usepackage[left=2.5cm,right=2.5cm,top=2.5cm,bottom=2.5cm]{geometry}
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{\textbf{Experiment 08}}
\fancyhead[R]{By Saksham}
\fancyfoot[L]{GNDEC}
\fancyfoot[R]{URN: 2435157}
\renewcommand{\headrulewidth}{1pt}
\renewcommand{\footrulewidth}{1pt}
\setlength{\headheight}{15pt}
\setlength{\headsep}{25pt}
\setlength{\footskip}{30pt}
\begin{document}
\title{\LARGE \textbf{Experiment 08: Demonstration of Subfigures in \LaTeX}}
\author{SAKSHAM}
\date{}
\maketitle
\thispagestyle{fancy}
\vspace{6pt}
\begin{center}
{\large Understanding the use of subfigure environment for arranging images side
by-side} \end{center}
\vspace{12pt}
\noindent
\textbf{Objective:}
The aim of this experiment is to demonstrate how multiple related images can be
arranged neatly within a single figure environment using the \texttt{subfigure}
package in \LaTeX. This approach helps in presenting visual comparisons and
grouped graphical outputs more effectively.
\vspace{6pt}
\textbf{Description:}
In this experiment, three different mathematical graphs are displayed side-by-side.
Each graph has its own caption and label, making it easy to compare their shapes
and behaviors. Subfigures are especially useful in documentation, reports, and
research papers where grouped visual information is required.
\section*{Graphs Included}
\begin{itemize}
\item \textbf{\$y = x\$} — A straight line representing a linear relationship.
\item \textbf{\$y = 3\sin x\$} — A sine wave with an amplitude of 3, showing periodic
oscillations.
\item \textbf{\$y = \frac{5}{x}\$} — A rational function representing a hyperbola with
```

```

inverse variation.
\end{itemize}
\section*{Subfigure Demo}
\begin{figure}[h]
\centering
\begin{subfigure}[b]{0.3\textwidth}
\centering
\includegraphics[width=\textwidth]{graph1.png}
\caption{$y = x$}
\label{fig:y_equals_x}
\end{subfigure}
\hfill
\begin{subfigure}[b]{0.3\textwidth}
\centering
\includegraphics[width=\textwidth]{graph2.png}
\caption{$y = 3\sin x$}
\label{fig:three_sin_x}
\end{subfigure}
\hfill
\begin{subfigure}[b]{0.3\textwidth}
\centering
\includegraphics[width=\textwidth]{graph3.png}
\caption{$y = \frac{5}{x}$}
\label{fig:five_over_x}
\end{subfigure}
\caption{Three mathematical graphs arranged side-by-side using subfigures}
\label{fig:three_graphs}
\end{figure}
\end{document}

```


AIM: Develop a LaTeX script to create a document that consists of two mathematical equations.

SCRIPT:

```

\documentclass[12pt,a4paper]{article}
\usepackage{geometry} \usepackage{amsmath, amssymb} \usepackage{fancyhdr}
\geometry{margin=1in}
\pagestyle{fancy} % for header and footer
\fancyhead[R]{By Saksham} % clear all header fields
\fancyhead[L]{\textbf{Experiment 09}}
\fancyfoot{} % clear all footer fields
\fancyfoot[L]{GNDEC} % Left footer
\fancyfoot[R]{URN: 2435157} % Right footer
\renewcommand{\footrulewidth}{1pt}
\title{\textbf{Mathematical Equations in LaTeX}}
\date{}
\begin{document}
\maketitle
\thispagestyle{fancy} % Show header/footer on title page
\section*{Mathematical Equations}
This document demonstrates two mathematical equations with brief explanations
written in \LaTeX.
\subsection*{1. Euler's Formula}
Euler's formula is a fundamental equation in complex analysis. It beautifully connects
the exponential function with the trigonometric functions sine and cosine:
\begin{equation}
e^{i\theta} = \cos\theta + i\sin\theta
\end{equation}
= This equation shows how complex exponentials can represent rotations in the
complex plane. It also forms the basis of Euler's identity, which is often called the
most elegant equation in mathematics.
\vspace{0.3cm}
\subsection*{2. Quadratic Formula}
The quadratic formula provides the solutions to the quadratic equation  $(ax^2 + bx + c = 0)$ . It is derived by completing the square and is used to find the roots of any
second
degree polynomial:
\begin{equation}
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{equation}
This formula is widely used in algebra, physics, and engineering to find real or
complex solutions. The discriminant  $(b^2 - 4ac)$  helps determine whether the roots
are real, equal, or complex.
\end{document}

```

AIM: Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

SCRIPT:

```
\documentclass{article}
\usepackage{geometry} \geometry{margin=1in} \usepackage{fancyhdr}
\pagestyle{fancy} \fancyhf{}
\fancyhead[R]{By Saksham}
\fancyhead[L]{Experiment 10}
\fancyfoot[L]{GNDEC}
\fancyfoot[R]{URN: 2435157}
\renewcommand{\headrulewidth}{1pt}
\renewcommand{\footrulewidth}{1pt}
\newtheorem{theorem}{Theorem}[section]
\newtheorem{definition}[theorem]{Definition}
\newtheorem{corollary}[theorem]{Corollary}
\newtheorem{lemma}[theorem]{Lemma}
\title{\LARGE \textbf{Theorem EnvironmentsX}}
\author{SAKSHAM}
\date{}
\begin{document}
\maketitle
\thispagestyle{fancy}
\section{Introduction}
\large
This document demonstrates how to present mathematical statements using LaTeX's
theorem environments.
\section{Main Results}
\large
\begin{theorem}[Pythagorean Theorem]
In a right-angled triangle, the square of the hypotenuse equals the sum of the
squares of the other two sides:
\begin{equation} a^2 + b^2 = c^2
\label{eq:pythagoras}
\end{equation}
\end{theorem}
\begin{proof}
\large
Let the triangle have sides  $a$ ,  $b$ , and hypotenuse  $c$ . By constructing squares
on each side and comparing areas, we find that the area of the square on the
hypotenuse equals the sum of the areas on the other two sides.
Equation~\ref{eq:pythagoras} confirms this relationship.
\renewcommand{\qedsymbol}{}
\end{proof}
\begin{definition}\large
```

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

`\end{definition}`

`\begin{definition}[Fibration]\large`

A fibration is a continuous map between topological spaces that satisfies the homotopy lifting property.

`\end{definition}`

`\begin{corollary}\large`

The number 2 is the only even prime number.

`\end{corollary}`

`\begin{lemma}\large`

If a divides b and b divides c , then a divides c .

`\end{lemma}`

`\section{Conclusion}\large`

We have illustrated the use of theorem environments in LaTeX with proper numbering, structure, and equation labeling.

`\end{document}`

AIM: Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section.

SCRIPT:

```
\documentclass{article} \usepackage{cite} \usepackage{fancyhdr}
\pagestyle{fancy} \fancyhf{} \fancyhead[R]{By Saksham}
\fancyhead[L]{Experiment 11}
\fancyfoot[L]{GNDEC}
\fancyfoot[R]{URN: 2435157}
\renewcommand{\footrulewidth}{0.4pt}
\begin{document}
\title{Microbial Cell Surfaces}
\author{SAKSHAM}
\date{}
\maketitle
\thispagestyle{fancy}
\section{Introduction}
Understanding the functions of microbial cell surfaces requires knowledge of their
structural and physical properties\cite{dufrene2002atomic}. Electron microscopy has
long been recognized as a key technique in microbiology to elucidate cell surface
ultra structure\cite{engel1999atomic}. An exciting achievement has been the
development of cryotechniques, which allow high-resolution imaging of cell
structures in conditions close to the native state\cite{franz2008atomic}. Yet direct
observation in aqueous solution remained impossible.
\section{Methods and Techniques}
Because of the small size of microorganisms, the physical properties of their
surfaces have been difficult to study\cite{marrese2017atomic}. Quantitative and
qualitative information on physical properties can be obtained by electron
microscopy techniques, X-ray photoelectron spectroscopy, infrared spectroscopy,
contact, and electrophoretic mobility measurements
\cite{altman2015noncontact}.
\cite{author2023techniques,researcher2022approach,scientist202
Recentadvances
methodology,developer2020innovation,expert2019analysis,analyst2018study} have
expanded our understanding of these properties.
\bibliographystyle{plain}
\begin{thebibliography}{10}
\bibitem{dufrene2002atomic}
Dufrene, Y. F. (2002). Atomic force microscopy, a powerful tool in microbiology.
Journal of Bacteriology, 184(19), 5205-5213.
\bibitem{engel1999atomic}
Engel, A., & Muller, D. J. (1999). Observing single biomolecules at work with the
atomic force microscope. Nature Structural Biology, 6(10), 811-817.
\bibitem{franz2008atomic}
```

Franz, C. M., & Muller, D. J. (2008). Analyzing focal adhesion structure by atomic force microscopy. *Journal of Cell Science*, 121(11), 1771-1780.

\bibitem{marrese2017atomic}

Marrese, M., Guarino, V., & Ambrosio, L. (2017). Atomic force microscopy: a powerful tool in biomedicine. *Biomedicine & Pharmacotherapy*, 93, 1023-1036.

\bibitem{altman2015noncontact}

Altman, M., Lee, J. R., & Tranchina, D. (2015). Noncontact atomic force microscopy: a tool for biological imaging. *Science Advances*, 1(8), e1500189.

\bibitem{author2023techniques}

Author, A. (2023). Advanced Techniques in Microbial Imaging. *Journal of Advanced Techniques*, 45(12), 123-134.

\bibitem{researcher2022approach}

Researcher, B. (2022). New Approaches to Microbial Surface Studies. *Microbial Research Journal*, 38(7), 678-689.

\bibitem{scientist2021methodology}

Scientist, C. (2021). Methodology in Microbial Surface Analysis. *Analytical Methods*, 55(9), 345-356.

\bibitem{developer2020innovation}

Developer, D. (2020). Innovations in Microscopy for Microbiology. *Microscopy Innovations Journal*, 30(5), 567-578.

\bibitem{expert2019analysis}

Expert, E. (2019). Analysis of Microbial Surfaces: New Insights. *Microbial Analysis*, 24(4), 234-245.

\bibitem{analyst2018study}

Analyst, F. (2018). Study of Microbial Surface Properties. *Journal of Microbial Studies*, 18(3), 145-157.

\end{thebibliography}

\end{document}

DATE: 10.11.2025

EXPERIMENT NO: 12

AIM: Develop a LaTeX script to create a simple report and article by using suitable commands and formats of Major/Minor project report.

SCRIPT:

```
\documentclass[a4paper,12pt]{report} \usepackage[utf8]{inputenc}
\usepackage{amsmath} \usepackage{amsfonts}
\usepackage{amssymb} \usepackage{graphicx}
\usepackage[left=3cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{fancyhdr} \usepackage{lipsum} % Header and footer
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{Experiment 12}
\fancyfoot[R]{\thepage}
\fancyfoot[L]{GNDEC}
\begin{document}
\begin{titlepage}
\centering
{\fontsize{24pt}{28pt}\selectfont \textbf{SEMINAR AND TECHNICAL REPORT
WRITING}}\par
\vspace{1cm}
{\fontsize{12pt}{14pt}\selectfont {SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE DEGREE OF}}\par
\vspace{0.5cm}
{\fontsize{14pt}{16pt}\selectfont \textbf{BACHELOR OF TECHNOLOGY}}\par
{\fontsize{14pt}{16pt}\selectfont
(Computer Science and Engineering))\par
\vspace{2cm}
\includegraphics[width=5cm]{GNDEC.png}\par\vspace{2cm}
\begin{center}
\vspace{1em}
\begin{minipage}[t]{0.46\textwidth}
\raggedright
\textbf{SUBMITTED BY}\[0.6em]
SAKSHAM\[0.4em]
URN: 2435157\[0.4em]
Section: CSE (E1)
\end{minipage}
\hfill
\begin{minipage}[t]{0.46\textwidth}
\raggedleft
\textbf{SUBMITTED TO}\[0.6em]
Dr. Amandeep Kaur Sohal\[0.4em]
Assistant Professor (CSE)
\end{minipage}
\end{center}
\end{titlepage}
\end{document}
```

```

\vfll
{\fontsize{14pt}{16pt}\selectfont \textbf{DEPARTMENT OF COMPUTER SCIENCE
AND ENGINEERING}}\par
{\fontsize{14pt}{16pt}\selectfont \textbf{GURU NANAK DEV ENGINEERING
COLLEGE}}\par
{\fontsize{14pt}{16pt}\selectfont \textbf{LUDHIANA, 141006}}\par
\vspace{0.5cm}
\end{titlepage}
\begin{abstract}
This is a brief summary of the report. It outlines the main objectives, methods, and
conclusions of the project. \lipsum[1]
\end{abstract}
\tableofcontents
\newpage
\chapter{Introduction}
\section{Background}
\lipsum[1]
\section{Objectives}
\lipsum[2]
\newpage
\chapter{Methodology}
\section{Data Collection}
\lipsum[1]
\section{Analysis}
\lipsum[1]
\newpage
\chapter{Results}
\section{Findings}
\lipsum[1]
\newpage
\chapter{Conclusion}
\lipsum[1]
\newpage
\begin{thebibliography}{9}
\bibitem{latexcompanion}
Leslie Lamport,
\textit{LaTeX: A Document Preparation System}, Addison
Wesley, 1986.
\bibitem{einstein}
Albert Einstein,
\textit{Relativity: The Special and the General Theory}, H.
Holt and Company, 1920.
\end{thebibliography}
\end{document}

```

AIM: Markdown and Pandoc

- **Introduction to .md format and Markdown syntax.**
- **Use cases of Markdown in documentation and technical writing.**
- **Conversion between LaTeX and Markdown using Pandoc.**

Introduction

Markdown is a lightweight markup language created by John Gruber in 2004, designed to make text formatting simple and readable. Files written in Markdown use the .md or .markdown extension. The main goal of Markdown is to allow users to write content in plain text that can easily be converted into rich formats like HTML, PDF, DOCX, or LaTeX. It is widely used for documentation, technical writing, blogging, and collaborative projects. Markdown files are human-readable and version-control friendly, making them ideal for developers and writers who prefer simplicity and portability over complex formatting interfaces.

Markdown Syntax and Structure

Markdown uses plain text symbols for formatting. Some common syntax elements include:

Purpose	Markdown Syntax	Rendered Output
Heading	# Heading 1 ## Heading 2	Heading 1 Heading 2
Bold	**bold text**	bold text
Italic	<i>*italic text*</i>	<i>Italic text</i>
List	- Item 1 - Item 2	• Item 1 • Item 2
Numbered list	1. Step 1 2. Step 2	1. Step 1 2. Step 2
Link	[OpenAI](https://openai.com)	OpenAI
Image	![Alt text](image.png)	(Display image)
Code	`code snippet`	Code snippet
Blockquote	> Quotation	> Quotation

Markdown focuses on content clarity instead of layout control. This makes it easy to learn and use even for beginners.

Use Cases of Markdown in Documentation and Technical Writing

- **Project Documentation**
 - Used for README.md files in GitHub or GitLab repositories.
 - Helps developers explain installation, usage, and contribution guidelines clearly.
- **Technical Writing**
 - Used for writing software manuals, API documentation, and developer guides.
 - Easier to version control compared to formatted documents.
- **Blogs and Websites**
 - Platforms like Jekyll, Hugo, and Hexo use Markdown for content creation.
 - Supports static site generation with easy customization.
- **Academic Writing**
 - Markdown can include citations, equations, and references (when combined with Pandoc).
 - Converts easily to LaTeX or PDF formats.
- **Collaborative Writing**
 - Tools like GitHub, Notion, and Obsidian support Markdown for team documentation and note-taking.
 - Allows easy sharing, reviewing, and formatting across platforms.

Conversion Between LaTeX and Markdown Using Pandoc

Pandoc: Pandoc is an open-source document conversion tool developed by John MacFarlane in 2006. It serves as a universal converter that allows smooth transformation between various document formats such as Markdown, LaTeX, HTML, PDF, and DOCX. It is often used in academic and research environments to integrate the simplicity of Markdown with the powerful formatting capabilities of LaTeX.

Converting Markdown to LaTeX

When a document is written in Markdown (.md) format and needs to be converted into a LaTeX (.tex) file, Pandoc can perform this task efficiently using the following command:

“pandoc file.md -o output.tex”

This command reads the Markdown source file file.md and generates a LaTeX-formatted file named output.tex. The resulting LaTeX file can then be compiled using any LaTeX editor to produce a PDF document or further customized for advanced formatting and styling.

Advantages of Using Pandoc for Conversion

- **Cross-format compatibility:** Converts between multiple file types (Markdown, LaTeX, PDF, DOCX, HTML).
- **Automation:** Useful for batch processing or academic publishing pipelines.
- **Consistency:** Ensures uniform formatting across different document types.
- **Integration:** Works well with bibliography tools like BibTeX and citation managers.
- **Simplicity:** Eliminates the need to rewrite content when changing output format.